Zaiqian "Ting" Chen

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Education

2024–Present **PhD, Statistics**, *Columbia University*, New York, NY.

2022–2024 **MS, Computer Science**, *Columbia University*, New York, NY, GPA: *3.99/4.00*.

2017–2021 **BS, Computer Science (honors), Mathematical Economics, Summa Cum Laude**, *University of Richmond*, Richmond, VA, GPA: *3.94/4.00*.

Fall 2019 **Visiting Student**, *University of Edinburgh*, Edinburgh, Scotland, GPA: *4.0/4.0* (converted).

Peer Reviewed Publications

ML4LMS Workshop @ ICML 2024 S. Chang, **Z. Chen**, B. Dumitrascu, D. A. Knowles, *CellFlows: Inferring Splicing Kinetics from Latent and Mechanistic Cellular Dynamics* in ICML'24 Workshop ML for Life and Material Science: From Theory to Industry Applications. 2024.

EMNLP 2022 **Z. Chen** D. Verdi do Amarante, J. Donaldson, Y. Jo and J. Park, *Argument Mining for Review Helpfulness Prediction* in Proceedings of the Empirical Methods on Natural Language Processing (EMNLP). 2022.

DSAA 2021 **Z. Chen** and J. Park, *Analyzing Cultural Assimilation through the Lens of Yelp Restaurant Reviews* at the 8th IEEE International Conference on Data Science and Advanced Analytics (DSAA 2021).

Research Experience

2022–2024 **Graduate Research Assistant**, *Department of Computer Science: Dr. David Knowles*, Columbia University, New York Genome Center.

- Developed systems to improve experimental design of massively parallel reporter assays (MPRAs) in order to better study the sequence determinants of alternative splicing
- Utilized deep generative models, convolutional neural networks, and conditional neural processes in order to more efficiently learn splicing outcomes from DNA sequences
- Trained DNA languages models with ATAC-Seq annotations in order to understand variant effects on chromatin accessibility

2020–2021 **Honors Thesis**, *Department of Computer Science: Dr. Joonsuk Park*, University of Richmond.

- Developed methods for graph-based argument mining using large language models, transfer learning, and dependency parsing methods
- Utilized annotations from argument mining model to improve performance on helpfulness prediction of Amazon product reviews
- Implemented methods in PyTorch and Huggingface's `transformers` library
- Project culminated into a EMNLP 2022 publication

Spring, Summer 2020 **Research Assistant**, *Department of Computer Science: Dr. Joonsuk Park*, University of Richmond.

- Developed probabilistic graphical models to perform topic modeling on online review data for cultural analysis
- Implemented graphical models with collapsed Gibbs Sampling in Python and Cython
- Project culminated into a DSAA 2021 publication

Summer 2019 **Summer Research Fellowship**, *Department of Computer Science: Dr. Prateek Bhakta*, University of Richmond.

- Developed algorithms towards fair and robust liver donation allocation strategies, focusing on maximum-flow algorithms and their properties
- Programmed simulations in Python and C++ to test developed algorithms
- Scrapped and cleaned medical and geographical data from publicly available sources for use in simulations

2018–2019 **Research Assistant**, *Department of Psychology: Dr. Cindy Bukach's Lab*, University of Richmond.

- Programmed experiments using Neurobehavioral Systems Presentation, Codex Superlab, and Python which investigated the Other Race Effect (ORE) and the Angry Black Bias
- Performed data cleaning and analysis using Python and R
- Created interactive web simulations using HYPE

Work Experience/Service

Spring 2025 **Teaching Assistant - Statistical Machine Learning**, *Department of Statistics*, Columbia University.

Fall 2024 **Teaching Assistant - Advanced Machine Learning**, *Department of Statistics*, Columbia University.

2021-2024 **Machine Learning Software Engineer**, *CarMax*.

- Built and maintained personalized recommendations systems as well as customer segmentation systems for search engine optimization
- Built microservices and common Python libraries in order to support data sourcing and model orchestration
- Built CI/CD pipelines in Azure DevOps in order to deploy cloud resources to Azure
- Trained, deployed and served model predictions through AzureML, Apache Spark, and Databricks

Fall 2023 **Reviewer**, *Nature Machine Intelligence*.

Spring 2019, Spring 2021 **Grading Assistant - Data Structures, Computer Systems and Organization**, *Department of Computer Science*, University of Richmond.

Spring 2019 **Grading Assistant - Linear Algebra**, *Department of Mathematics*, University of Richmond.

Technical Skills

Relevant Coursework: Foundations of Graphical Models, Machine Learning in Genomics, Deep and Generalized Probabilistic Models, Neural Networks and Deep Learning, Geometric Data Analysis, Interpretable Machine Learning for Inverse Problems in Biology

Software: PyTorch, TensorFlow, Python, LaTeX, Apache Spark, C++, Microsoft Azure Cloud Services, Google Cloud Platform (GCP), Azure Cosmos DB, Azure DevOps, Azure ML

Honors

2021 **Phi Beta Kappa**, University of Richmond

2017–2021 **University of Richmond Science Scholar Award:** Full tuition merit scholarship for 4 years

2017–2021 **National Merit Scholarship**

2017 **National AP Scholar**

2016 **Most Interesting Hack** at Vassar College Hackathon